Questions and MCQ
Step 1 : Calculate Cash Flows

1) Which of the following statements is FALSE?
A) Sales will ultimately decline as the product nears obsolescence or faces increased competition.
B) Managers sometimes continue to invest in a project that has a negative NPV because they have already invested a large amount in the project and feel that by not continuing it, the prior investment will be wasted.
C) With straight-line depreciation the asset's cost is divided equally over its life.
D) A project's unlevered net income is equal to its incremental revenues less costs and depreciation, evaluated on a pre-tax basis.
Answer: D
Explanation: A project's unlevered net income is equal to its incremental revenues less costs and depreciation, evaluated on an after-tax basis.

2) Which of the following statements is FALSE?
A) The ultimate goal in capital budgeting is to determine the effect on the firm's cash flows of the decision to take a particular project..
B) To the extent that overhead costs are fixed and will be incurred in any case, they are incremental to the project and should be included in the capital budgeting analysis.
C) Unlevered Net Income = (Revenue - Costs - Depreciation) × (1 - τc).
D) Earnings are not cash flows.
Answer: B
Explanation: To the extent that overhead costs are fixed and will be incurred in any case, they are NOT incremental to the project and should NOT be included in the capital budgeting analysis.

3) The value of currently unused warehouse space that will be used as part of a new capital budgeting project is:
A) an opportunity cost.
B) irrelevant to the investment decision.
C) an overhead expense.
D) a sunk cost.
Answer: A

Use the information for the question(s) below.

Glucose Scan Incorporated (GSI) currently sells its latest glucose monitor, the Glucoscan 3000, to diabetic patients for $129. GSI plans on lowering their price next year to $99 per unit. The cost of goods sold for each Glucoscan unit is $50, and GSI expects to sell 100,000 units over the next year.

4) Suppose that if GSI drops the price on the Glucoscan 3000 immediately, it can increase sales over the next year by 30% to 130,000 units. The incremental impact of this price drop on the firm's EBIT is closest to:

A) a decline of $1.5 million.
B) an increase of $1.5 million.
C) a decline of $2.4 million.
D) an increase of $2.4 million.
Answer: A
Explanation: Without price cut = 100,000 units × ($129 - 50) = $7,900,000
With price cut = 130,000 units × ($99 - 50) = $6,370,000
So, incremental = $6,370,000 - $7,900,000 = -$1,530,000

5) What is a sunk cost? Should it be included in the incremental cash flows for a project? Why or why not?
Answer: A sunk cost is any unrecoverable cost for which the firm is already liable. Sunk costs will have to be paid regardless of the decision whether or not to proceed with the project. Therefore, sunk costs are not incremental with respect to the current decision regarding the project and should not be included in its analysis.

6) Your firm is considering building a new office complex. Your firm already owns land suitable for the new complex. The current book value of the land is $100,000. However a commercial real estate agent has informed you that an outside buyer is interested in purchasing this land and would be willing to pay $650,000 for it. When calculating the NPV of your new office complex, ignoring taxes, the appropriate incremental cash flow for the use of this land is:
A) $650,000.
B) $0.
C) $100,000.
D) $750,000.
Answer: A
Explanation: It is appropriate to use the market value. If taxes are included, the value would be the after-tax value of the land.

Use the information for the question(s) below.

The Sisyphean Corporation is considering investing in a new cane manufacturing machine that has an estimated life of three years. The cost of the machine is $30,000 and the machine will be depreciated straight line over its three-year life to a residual value of $0.

The cane manufacturing machine will result in sales of 2000 canes in year 1. Sales are estimated to grow by 10% per year each year through year three. The price per cane that Sisyphean will charge its customers is $18 each and is to remain constant. The canes have a manufacturing cost of $9 each.

Installation of the machine and the resulting increase in manufacturing capacity will require an increase in various net working capital accounts. It is estimated that the Sisyphean Corporation needs to hold 2% of its annual sales in cash, 4% of its annual sales in accounts receivable, 9% of its annual sales in inventory, and 5% of its annual sales in accounts payable. The firm is in the 21% tax bracket, and has a cost of capital of 10%.

7) The required net working capital in the first year for the Sisyphean Corporation's project is closest to:

A) $3600.
B) $3960.
C) $2880.
D) $5400.
Answer: A
Explanation:

Networking Capital Forecast ($)

Year	**1**	**2**	**3**
Units	2000	2200	2420
Sales (units × $18)	36,000	39,600	43,560
Cash (2% of sales)	720	792	871.2
Accounts Receivable (4% of sales)	1440	1584	1742.4
Inventory (9% of sales)	3240	3564	3920.4
Accounts Payable (5% of sales)	1800	1980	2178
NWC (Cash + Inventory+ AR - AP)	3600	3960	4356

Use the information for the question(s) below.

Epiphany Industries is considering a new capital budgeting project that will last for three years. Epiphany plans on using a cost of capital of 12% to evaluate this project. Based on extensive research, it has prepared the following incremental cash flow projections:

Year	**0**	**1**	**2**	**3**
Sales (Revenues in $)		100,000	100,000	100,000
- Cost of Goods Sold (50% of Sales)		50,000	50,000	50,000
- Depreciation		30,000	30,000	30,000
= EBIT		20,000	20,000	20,000
- Taxes (21%)		4200	4200	4200
= Unlevered Net Income		15,800	15,800	15,800
+ Depreciation		30,000	30,000	30,000
+ Changes to Working Capital		-5000	-5000	10,000
- Capital Expenditures	-90,000			

8) The free cash flow for the first year of Epiphany's project is closest to:
A) $43,000.
B) $25,000.
C) $40,800.
D) $45,000.
Answer: C
Explanation:

Year	**0**	**1**	**2**	**3**
Sales (Revenues)		100,000	100,000	100,000
- Cost of Goods Sold (50% of Sales)		50,000	50,000	50,000
- Depreciation		30,000	30,000	30,000
= EBIT		20,000	20,000	20,000
- Taxes (21%)		4200	4200	4200
= Unlevered Net Income		15,800	15,800	15,800

+ Depreciation		30,000	30,000	30,000
+ Changes to Working Capital		-5000	-5000	10,000
- Capital Expenditures	-90,000			
= Free Cash Flow	-90,000	40,800	40,800	55,800

PV of FCF ($FCF/(1 + I)n$	-90,000	36,428	32,526	39,717
Discount Rate	0.12			
NPV = 18,671				
IRR = 23.09%				

Use the following information to answer the question(s) below.

Two years ago, Krusty Krab Restaurant purchased a grill for $50,000. The owner, Eugene Krabs, has learned that a new grill is available that will cook Krabby Patties twice as fast as the existing grill. This new grill can be purchased for $80,000 and would be depreciated straight line over 8 years, after which it would have no salvage value. Eugene Krab expects that the new grill will produce EBITDA of $50,000 per year for the next eight years while the existing grill produces EBITDA of only $35,000 per year. The current grill is being depreciated straight line over its useful life of 10 years after which it will have no salvage value. All other operating expenses are identical for both grills. The existing grill can be sold to another restaurant now for $30,000. Krusty Krab's tax rate is 21%.

9) The incremental cash flow that Krusty Krab will incur today (Year 0) if they elect to upgrade to the new grill is closest to:

A) -$80,000.
B) -$50,000.
C) -$47,900.
D) +$30,000.
Answer: C
Explanation: CF0 = -$80,000 + $30,000 + .21($40,000 - $30,000) [tax write-off old grill sold at loss] = -$47,900

Casa Grande Farms is considering purchasing multiple tractors for a total purchase price of $540,000. These tractors are expected to generate EBITDA of $250,000 for each of the next three years. Casa Grande Farms has a 21% tax rate and has a cost of capital of 10%.

10) Assuming that Casa Grande Farms depreciates these tractors straight line over the three-year life, then the annual depreciation tax shield in year 2 is closest to:

A) $37,800.
B) $80,000.
C) $84,000.
D) $117,000.
Answer: A
Explanation: Depreciation Tax Shield = ($540,000/3) × .21 = $37,800

Use the following information to answer the question(s) below.

Really Big Conglomerate (RBC) is considering acquiring POP, Inc., a smaller unsuccessful Internet firm.

POP has outstanding tax loss carryforwards of $320 million from losses over the past six years. RBC has pre-tax income of $100 million per year, a cost of capital of 10%, and pays 21% in taxes. The Tax Cuts and Jobs Act of 2017 will limit RBC's ability to write off the carryforwards to 80% of RBC's annual pre-tax income.

11) If RBC acquires POP, in what year will RBC completely use up the tax loss carryforward?
A) 2 years
B) 3 years
C) 4 years
D) 5 years
Answer: C
Explanation: The tax shield will last for $320 million/($100 million × 0.80) = 4.0 years.

12) If RBC acquires POP, then the NPV of POP tax loss carryforwards to RBC is closest to:
A) $53 million.
B) $236 million.
C) $262 million.
D) $320 million.
Answer: A
Explanation: The total of $320 million in tax loss carryforwards will offset income of $80 million in years 1-4. The tax savings will be 21% of these amounts, so the NPV of the tax loss carryforwards in millions is $16.8 million/(1.10)1 + $16.8 million/(1.10)2 + $16.8 million(1.10)3 + $16.8 million/(1.10)4 = $53.25 million

13) An exploration of the effect on NPV of changing multiple project parameters is called:
A) scenario analysis.
B) IRR analysis.
C) accounting break-even analysis.
D) sensitivity analysis.
Answer: A

14) What is sensitivity analysis?
Answer: Sensitivity analysis breaks the NPV calculation into its component assumptions and shows how the NPV varies as each of the underlying assumptions change. Sensitivity analysis allows us to explore the effects of errors in the estimated inputs to the NPV calculations and reveals which aspects of the project are most critical when actually managing the project.